Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	**AI-Based Adaptive Fitness and Workout Planner**	
S. No.	**University ID**	**Name**
1	2420030498	K. Sanuhya
2	2420030516	R. Sarvajith Varma
3	2420030678	G. Gopi Krishna
Name of the Guide		Chandra Shekhar Rendla

Abstract

The AI-Based Adaptive Fitness and Workout Planner is a personalized fitness application designed to create and continuously improve workout plans according to individual user needs and performance. The system collects information such as fitness goals, fitness level, preferred workout type, available workout time, and previous workout performance. Using Artificial Intelligence and recommendation techniques, it analyses the user's progress and generates suitable workout plans. It can also consider factors such as workout history, exercise preferences, and achieved fitness targets to provide more relevant and personalized recommendations. The system is adaptive because it modifies the workout intensity, duration, exercises, and recommendations based on the user's performance and feedback. For example, if a user finds a workout too difficult or shows improved performance, the system can adjust the next workout accordingly. The application also provides progress tracking and personalized recommendations to help users maintain consistent fitness routines. The proposed system aims to provide a flexible, user-friendly, and intelligent fitness planning solution that adapts to changing user requirements and supports effective workout management.

Signature of the Guide HOD